

EXPERIMENT 5

```
mysql> create database sailors;
Query OK, 1 row affected (0.02 sec)
```

```
mysql> use sailors;
Database changed
```

```
mysql> CREATE TABLE Sailors (
->     sid INT PRIMARY KEY,
->     sname VARCHAR(30),
->     rating INT,
->     age DECIMAL(4,1)
-> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO Sailors VALUES
-> (1,'Bob',7,25.5),
-> (2,'Alice',5,22.0),
-> (3,'Charlie',7,28.0),
-> (4,'David',8,20.0),
-> (5,'Eva',5,24.5),
-> (6,'Frank',6,19.0),
-> (7,'Grace',8,21.5),
-> (8,'Helen',6,26.0);
Query OK, 8 rows affected (0.03 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Boats (
->     bid INT PRIMARY KEY,
->     bname VARCHAR(30),
->     color VARCHAR(20)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> INSERT INTO Boats VALUES
-> (101,'Sea Queen','Red'),
-> (102,'Blue Wave','Blue'),
-> (103,'Green Pearl','Green'),
-> (104,'Ocean King','Red'),
-> (105,'River Star','White');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE Reserves (
->     sid INT,
->     bid INT,
->     day DATE,
->     PRIMARY KEY (sid,bid,day),
->     FOREIGN KEY (sid) REFERENCES Sailors(sid),
->     FOREIGN KEY (bid) REFERENCES Boats(bid)
```

```
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
mysql> INSERT INTO Reserves VALUES
```

```
-> (1,101,'2024-01-10'),
```

```
-> (1,102,'2024-01-10'),
```

```
-> (2,103,'2024-02-05'),
```

```
-> (3,104,'2024-03-12'),
```

```
-> (3,105,'2024-03-12'),
```

```
-> (4,101,'2024-04-15'),
```

```
-> (5,102,'2024-05-20'),
```

```
-> (5,104,'2024-05-25'),
```

```
-> (6,103,'2024-06-18'),
```

```
-> (7,101,'2024-07-01'),
```

```
-> (8,105,'2024-07-10');
```

Query OK, 11 rows affected (0.02 sec)

Records: 11 Duplicates: 0 Warnings: 0

```
mysql> select * from reserves;
```

sid	bid	day
1	101	2024-01-10
4	101	2024-04-15
7	101	2024-07-01
1	102	2024-01-10
5	102	2024-05-20
2	103	2024-02-05
6	103	2024-06-18
3	104	2024-03-12
5	104	2024-05-25
3	105	2024-03-12
8	105	2024-07-10

11 rows in set (0.00 sec)

```
mysql> SELECT s.*
```

```
-> FROM Sailors s
```

```
-> JOIN Reserves r
```

```
-> ON s.sid = r.sid
```

```
-> WHERE r.bid = 101;
```

sid	sname	rating	age
1	Bob	7	25.5
4	David	8	20.0
7	Grace	8	21.5

3 rows in set (0.01 sec)

```
mysql> SELECT b.bname
-> FROM Sailors s
-> JOIN Reserves r
-> ON s.sid = r.sid
-> JOIN Boats b
-> ON r.bid = b.bid
-> WHERE s.sname = 'Bob';
```

```
+-----+
| bname |
+-----+
| Sea Queen |
| Blue Wave |
+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT s.sname
-> FROM Sailors s
-> JOIN Reserves r
-> ON s.sid = r.sid
-> JOIN Boats b
-> ON r.bid = b.bid
-> WHERE b.color = 'Red'
-> ORDER BY s.age ASC;
```

```
+-----+
| sname |
+-----+
| David |
| Grace |
| Eva   |
| Bob   |
| Charlie |
+-----+
```

5 rows in set (0.00 sec)

```
mysql> SELECT DISTINCT s.sname
-> FROM Sailors s
-> JOIN Reserves r
-> ON s.sid = r.sid;
```

```
+-----+
| sname |
+-----+
| Bob   |
| Alice |
| Charlie |
| David |
| Eva   |
| Frank |
| Grace |
| Helen |
+-----+
```

8 rows in set (0.00 sec)

```
mysql> SELECT DISTINCT s.sid, s.sname
-> FROM Sailors s
-> JOIN Reserves r1
-> ON s.sid = r1.sid
-> JOIN Reserves r2
-> ON r1.sid = r2.sid
-> AND r1.day = r2.day
-> AND r1.bid <> r2.bid;
```

sid	sname
1	Bob
3	Charlie

2 rows in set (0.00 sec)

```
mysql> SELECT DISTINCT r.sid
-> FROM Reserves r
-> JOIN Boats b
-> ON r.bid = b.bid
-> WHERE b.color = 'Red' OR b.color = 'Green';
```

sid
1
4
7
2
6
3
5

7 rows in set (0.00 sec)

```
mysql> SELECT sname, age
-> FROM Sailors
-> WHERE age = (
->     SELECT MIN(age)
->     FROM Sailors
-> );
```

sname	age
Frank	19.0

1 row in set (0.00 sec)

```
mysql> SELECT COUNT(DISTINCT sname) AS Total_Sailors
```

```
-> FROM Sailors;
```

```
+-----+
| Total_Sailors |
+-----+
|              8 |
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT rating, AVG(age) AS Average_Age
```

```
-> FROM Sailors
```

```
-> GROUP BY rating;
```

```
+-----+-----+
| rating | Average_Age |
+-----+-----+
|      7 |    26.75000 |
|      5 |    23.25000 |
|      8 |    20.75000 |
|      6 |    22.50000 |
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT rating, AVG(age) AS Average_Age
```

```
-> FROM Sailors
```

```
-> GROUP BY rating
```

```
-> HAVING COUNT(*) >= 2;
```

```
+-----+-----+
| rating | Average_Age |
+-----+-----+
|      7 |    26.75000 |
|      5 |    23.25000 |
|      8 |    20.75000 |
|      6 |    22.50000 |
+-----+-----+
```

```
4 rows in set (0.00 sec)
```

```
mysql>
```